

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_f9255361-69d\agent-  
a22df82ecda012c00.jsonl`  
- SHA-256: `7257adbe85e957566f2323c69df976670c946c1b8e8bf29f43ef4e782edde33c`  
- Source modified: `2026-06-22T09:36:03+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_f9255361-69d`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-22T09:32:46.028Z)

You are auditing ONE doc in the KhelSutra brand/site repo for staleness, as part of a "are all the docs up to date?" sweep.

Repo root: C:\Users\avidu\Projects\khelesutra (bash path: /c/Users/avidu/Projects/khelesutra)
The repo is on `master`, freshly synced to remote HEAD ? the latest merged PR is #94. Today is 2026-06-22.

DOC TO AUDIT: docs/PER_RALLY_SELECTION.md

Your job: read the doc IN FULL, extract every concrete, falsifiable claim, and VERIFY each against the LIVE repo ? do NOT trust the doc's own wording. Hard discipline rule: a claim is "correct" ONLY when you have file:line or command-output evidence for it; if you cannot verify it, mark verdict "unverifiable" ? never guess, never assume the doc is right. Re-check against actual code/git.

Kinds of claims to check:

- File / path / route references ("served at /capture", "see src/store.js", "public/og.png", "systemd khelesutra-site") -> does the path or route ACTUALLY exist now? Confirm with git ls-files / Read / the routing in site/server.js + site/src/index.js.
- PR numbers and counts ("PRs #1-#8 merged", "20 tests green", "6 languages", "3-step wizard", "P0..P5 done") -> verify against `git -C <root> log`, the actual test output/count, and the code.
- Status markers (DONE / pending / IN PROGRESS / DRAFT / "?" / "owner-gated" / "NEXT") -> still accurate vs the code & git history? Is something marked pending that is actually shipped, or vice-versa?
- Dates, "live at <url>", "merged in #N".
- Whether the doc describes something now SUPERSEDED or fully shipped -> is it an archive candidate (a tracker whose work is complete, or content duplicated by docs/archives/)? Note: docs/archives/FRIENDLY_AND_MULTILINGUAL.md already exists as the archived multilingual tracker.

Tools: Read the doc + every cited file; Grep/Glob to locate code; Bash for `git -C /c/Users/avidu/Projects/khelesutra log --oneline`, `... ls-files`, and greps. To check a test count, you may run the suite (`cd /c/Users/avidu/Projects/khelesutra/site && node --test` or `npm test`), but prefer cheap checks. Verify against master as it stands NOW.

Severity: high = a factually wrong claim a reader would act on (wrong path/route, a "DONE" that isn't, a wrong count/PR that misleads). medium = outdated-but-harmless (stale PR range, old "where we are", superseded framing). low = cosmetic.

Do NOT modify any file ? audit only. Return ONLY the structured object (doc, overallStatus, findings[], summary).

2. assistant (2026-06-22T09:32:48.695Z)

I'll audit the doc against the live repo. Let me start by reading the doc in full and syncing the repo state.

3. user (2026-06-22T09:32:50.184Z)

```
1      # Interactive per-rally selection ? reel builder (RallyPicker)
2
3      > **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ?
select ? compile ? download, on existing engine endpoints). **Validated live e2e against a REAL
engine (Gemini) via the in-UI bucket?reel CUJ ? `DRY_RUN_CUJ.md` S9, 2026-06-21** (supersedes
the original stub-engine verification). Design **APPROVED by owner 2026-06-20**. Downstream
milestones M1?M5 (multi-video auth, per-rally preview endpoint, corpus loop, localized NL)
remain owner-gated. . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** S7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally
**list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest
first") pre-selects into the **same** selection set; a sticky footer compiles the selection into
one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ?
`POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396,
307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most
important caveat: queries and chips are scoped to **length / count / order / shot-count only**;
shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be
faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the
**final compiled reel* can stream ? so MVP selects on text metadata, and visual preview/timeline
is a deliberate later milestone.
13
14     ## 1. Problem & goal
15     Today the alpha can detect rallies and stitch a reel, but reel-building is either a
**silent `stitching.min_shot_count` threshold** `[verified backend/main.py:362-383]` or a static
checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies`
while the API returns `play_segments`, so it renders nothing `[verified
backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-
video selection, not a fixed threshold**: a coach should say "give me the long rallies from this
session" and get exactly those, then fine-tune by hand.
16
17     **The concrete outcome ("good")**: A coach uploads a 40-minute session; the engine finds
60 rallies. They type **"top 15 longest", eyeball the auto-selected cards, untick two warm-up
rallies, name it **"Tuesday ? long points", and download a ~6-minute reel ? without learning any
threshold knob.
18
19     **Goal**: the smallest **honest** surface that turns engine-detected rallies into a user-
curated, downloadable reel, with both **fast** (query) and **precise** (checkbox) selection
sharing **one** state.
20
21     **Non-goals (alpha)**: no conversational LLM (`UI_PROPOSAL.md:66` defers it); no shot-
type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt
streaming; no multi-tenant auth (single-box alpha).
22
23     ## 2. Decisions locked
```

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24      | # | Decision | Source / date | Implication |
25      |---|-----|-----|-----|
26      | D1 | Dominant surface = selectable rally **list/grid** | Synthesis 2026-06-20 (Grid
scored 15, fit-to-data 4) | Scales to long matches via bulk+filter; shows honest fields only |
27      | D2 | Fast entry = **deterministic** client-side `parseQuery()` (no LLM) | `[verified
UI_PROPOSAL.md:66]` conversational QA deferred | Predictable, unit-testable, offline, zero API
cost |
28      | D3 | **One** shared `Set<segment_id>`, mutated by query + checkbox + (future) timeline
| Synthesis 2026-06-20 | No second compile path can diverge from what the user sees |
29      | D4 | Compile via existing `POST /api/compile {video_id, segment_ids, output_filename}`
| `[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher
`_merge_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |
30      | D5 | Query grammar (alpha) = length, count, order, **shot_count only** | `[verified
database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out
the rest |
31      | D6 | MVP backend = **zero new endpoints** | `[verified]` reuse map (§4) | Ships the
full loop on the already-shipped contract |
32      | D7 | Replace the silent `min_shot_count` cut with **explicit user selection** | Owner
framing + `[verified main.py:362-383]` | UI *warns* about the floor; it never hides why a rally
vanished |
33      | D8 | **Default selection = ON** (curate-by-removal) | Owner 2026-06-20 | Rallies load
all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX
assumes a non-empty start |
34      | D9 | **Preview-before-select is NOT a ship-blocker** for the MVP | Owner 2026-06-20 |
MVP selects on text metadata + previews the *final compiled reel*; the source/clip-stream
endpoint (per-rally preview) is the next milestone (P9), not MVP |
35      | D10 | **MVP is job-tethered** (carry `result.video_id`); no `GET /api/videos` | Owner
2026-06-20 | Smallest honest slice; multi-video home + identity scoping is P8 (gated) |
36      | D11 | `min_shot_count`: **warn, do NOT override** in the MVP | Owner 2026-06-20 | The
UI surfaces the floor; an explicit-selection override of the compile default is deferred (would
trigger the Launch Report convention) |
37
38      ## 3. Foundation ? what the engine gives us today `[verified this session]`
39      The whole design rests on the engine's real, already-shipped contract. Read against the
engine repo (`badminton-highlight-indexer`):
40
41      - **The rally store is `play_segments`** ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
42      - **Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}`** ? the
*only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py:331-340, database.py:427-475]`.
43      - **Building a reel = `POST /api/compile {video_id, segment_ids:[int],
output_filename}`** ? loads `videos.filepath`, filters all segments to the requested
`segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and
calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which **merges overlaps** then per-clip re-encodes ? concat ? one MP4
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.
44      - **Downloading = `GET /api/highlights/{video_id}/{filename}`** ? streams the compiled
MP4. Compile returns a **server-local filepath, NOT a URL** `[verified main.py:394]`, so the SPA
constructs this download URL itself. The compiled reel is the **only streamable video that
exists today** `[verified main.py:307-328]`.
45      - **The floor knob is readable** ? `GET /api/config` exposes `stitching.min_shot_count`
`[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.
46      - **Job entry** ? `GET /api/jobs/{job_id}` returns `result.video_id` `[verified
main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos
endpoint.
47
48      **Honest fields vs fabricated/roadmap (critical for the UI):**
49      - **Honest today** (default `default_segmenter: "gemini"`, per `config.json`
`[verified]`): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))` + `metadata.shot_count_confidence` (a
**string**, often `"Unknown"`) `[verified gemini.py:470-485]`. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.

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50     - **Fabricated ? do NOT surface:** the `motion` demo segmenter fills `server`/`receiver`
via `random.sample(...)` , a random `shot_count` , and random `events` (serve/smash/foul/winner)
`[verified motion.py:191-198]` . `query` returns these columns + `events` , but they are not
ground truth and must stay hidden.
51     - **Roadmap ? no column, no detector:** shot-type per shot, player identity, match/point
score, game/set grouping. None exist anywhere in the schema; do not consume them `[verified
database.py whole-schema 122-148]` .
52     - **Correction-loop columns exist but are dead:**
`candidate_segments.human_label/notes/confirmed_segment_id` are defined but no endpoint writes
them `[verified database.py:166-168]` ? that's the M4 corpus extension (engine PR-4/B6,
`UI_ALPHA_EXECUTION_PLAN.md`) .
53
54     ## 4. Design / architecture
55
56     **Reuse map (existing ? `[verified]` , no change for MVP):** `POST /api/query` (list) ?
`POST /api/compile` (stitch) ? `GET /api/highlights/{video_id}/{filename}` (download); `GET
/api/config` (floor) ; `GET /api/jobs/{id}` (carry `result.video_id`) . **All net-new code is the
SPA** in `khelesutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`) , plus a small
set of *optional* real-gap engine enablers (§4.3) ? none required for MVP.
57
58     **4.1 Components (net-new SPA)**
59     - **QueryBar + `parseQuery()`** ? a *pure, deterministic, unit-tested* string ?
`{filter, sort}` compiler over `duration` / `ordinal` / `shot_count` . Renders a **parsed-pill**
("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-
correctable; greys out shot-type/player/score tokens with an inline "not available yet" . No LLM,
no network call.
60     - **RallyList/Grid** ? one card per `play_segment` : ordinal (by `start_time`), `mm:ss`
start, duration (`end?start`), and a shot-count + confidence chip **only when
`metadata.shot_count` is present** (absent on detector paths that don't write it) . Checkbox per
card; the thumbnail area is a placeholder (no endpoint yet) .
61     - **useRallySelection()` hook** ? owns `selectedIds:Set<number>` , `displayOrder` ,
`apply(queryResult)` , `toggle(id)` , and bulk Select-all / Clear / Invert . Unit-tested (matches
the exec-plan's "API client + review-state logic" bar) .
62     - **SelectionFooter (sticky)** ? live "N clips / total M:SS" ; a **`min_shot_count`
warning** when a selected rally would be dropped at compile (read from `GET /api/config`) ; reel-
name input; one **Build** button.
63     - **CompileResult** ? <video src=GET /api/highlights/...> `preview of the *compiled
reel* + a download link.
64     - **TypedApiClient** ? `query` , `compile` , `buildDownloadUrl` ; the typed seam to the
engine.
65
66     **4.2 Data flow** `[verified anchors inline]`
67     Gemini segmenter (process time) writes `play_segments {id, start/end_time, duration,
metadata.shot_count/confidence/detector}` `[gemini.py:461-498]`
68     ? SPA `POST /api/query {video_id}` ? `{play_segments:[...]}` `[main.py:331-340]`
(**fix** the static UI's `data.rallies` read ? `play_segments` `[app.js:344]` )
69     ? normalize to `RallyCard[]` + one shared `selectedIds:Set` ;
`server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **not
surfaced** `[motion.py:191-198]`
70     ? all input modes mutate the **same** `selectedIds` : (a) `parseQuery()` predicate over
duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band
71     ? footer recomputes count + ?duration **client-side** , and warns for any selected rally
with `shot_count < stitching.min_shot_count` `[main.py:362-383]`
72     ? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds],
output_filename}` ? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]` , cuts
`time_pairs` from `videos.filepath` , re-encodes per clip ? concat ? one MP4 `[main.py:386-393,
compiler.py:303]`
73     ? compile returns a **server-local filepath, not a URL** `[main.py:394]` ? SPA builds
`GET /api/highlights/{video_id}/{output_filename}` to preview in <video> + download
`[main.py:307-328]` .
74
75     **4.3 Engine API additions (none for MVP; tagged against real gaps)**
76     | Endpoint | Purpose | Status |
77     |---|---|---|
78     | `POST /api/query` | List rallies (only path) ? fix SPA to read `play_segments` |

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**exists** |
79 | `POST /api/compile` | Stitch selected `segment_ids` ? MP4 | **exists** |
80 | `GET /api/highlights/{video_id}/{filename}` | Stream/download compiled reel |
**exists** |
81 | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82 | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83 | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify**
|
84 | `GET /api/health` | Engine-offline banner ping the SPA is specified to use
(`HOSTING_PLAN.md:42`) | **new** |
85 | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same
`{play_segments}` shape | **new** |
86 | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)`
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87 | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream
source/per-span video so a rally previews before selection ? neither exists | **new** |
88 | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still
(no thumbnail column; WASB frames are transient) | **new** |
89 | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91 *** Dead-ends / do-NOT:*** never surface `server`/`receiver`/`events` (motion-only
randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a
confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `Unknown`) and
`play_segments` has **no** numeric confidence column.
92
93 **4.4 Observability (first-class, per `[memory: feedback-launch-quality-bar]`).** Every
P3 user action emits a **structured JSON log line** (`web/src/lib/log.ts`) so a journey is
traceable from text to selection ? the SPA-side "runlog". Each query application generates **one
`cid`** shared by every line of that journey; the future `/api/compile` call (P5) reuses the
same `cid` to thread the trace into the engine. Events (all on the P3 path):
94
95 | event | level | when | key fields |
96 |---|---|---|---|
97 | `query.parse` | info | every apply attempt | `cid`, `source` (typed/example),
`recognized`, `filters`, `sort`, `limit`, `unavailable` ? the exact interpretation |
98 | `query.unavailable_request` | info | the query names a shot-type/player/score | `cid`,
`raw`, `unavailable[]` ? **demand signal** for roadmap features (what users ask for that we
can't yet do) |
99 | `query.skip` | info | nothing actionable parsed | `cid`, `raw`, `reason` |
100 | `query.apply` | info | a recognized query is applied | `cid`, counts
(`total`/`selected`), `selected_ids`, `dropped_unavailable` |
101 | `query.empty_result` | **warn** | recognized query matches **zero** rallies | `cid` +
full interpretation ? a looks-empty grid is **never silent** |
102 | `query.apply_failed` | **error** | the (pure) parser/selector unexpectedly throws |
`cid`, `raw`, `error` ? loud, never swallowed |
103 | `selection.bulk` | info | Select-all / Clear / Invert | `cid`, `op`, `result_count` |
104 | `rally.toggle` | debug | a card is added/removed | `id`, `action` |
105
106 No silent drops: unsupported concepts (`query.unavailable_request` +
`dropped_unavailable`) and empty results (`query.empty_result` warn) are always logged. Verified
live in-browser (events fire with correct `cid`/fields, zero console errors).
107
108 ## 5. Risk table
109 | Risk | Severity | Mitigation |
110 |---|---|---|
111 | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
112 | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
113 | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true

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scrubber until source streams |
114 | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
115 | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
116 | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
117 | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, $8) |
118
119 ## 6. Honest limits ? what this does NOT do
120 A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic
only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END**
weakness is a separate quality track, unaffected by this UI). "Conversational" here means a
**deterministic four-dimension parser**, not natural-language understanding. Nothing is
implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already
found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's
the M4 correction loop).
121
122 ## 7. Deliverables & progress tracker ? **source of truth**
123 Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master`
auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
124
125 | ID | Deliverable | Depends on | Gated? | Status | PR |
126 |----|-----|-----|-----|-----|----|
127 | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? |
[11](https://github.com/avidullu/khelsutra/pull/11) |
128 | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | [14](https://github.com/avidullu/khelsutra/pull/14) |
129 | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | [21](https://github.com/avidullu/khelsutra/pull/21) |
130 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | (this PR) |
131 | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ?
| (this PR) |
132 | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | (this PR) |
133 | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
134 | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
135 | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |
136 | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
137 | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
138
139 **Milestone rollup:** **M0/MVP = P1?P5 ? COMPLETE (2026-06-20)** (pick?reel on existing
API, zero new engine endpoints) . **M1 = P6?P7** (RESTful + reliability polish) . **M2 = P8**
(multi-video home) . **M3 = P9** (per-rally preview/timeline) . **M4 = P10** (corpus correction
loop) . **M5** = swap `parseQuery()` for localized NL behind the same seam (gated on shot/player
data landing).
140
141 ## 8. Open questions ? owner / external
142 **Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in $2:**
143 1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
144 2. ~~Is preview-before-select a ship-blocker?~~ ? **No** ; text-metadata MVP, preview is
P9 (D9).
145 3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
146 4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).
147
148 **Still open (nice-to-knows, non-blocking):**

```

149 5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is explicitly deferred (`UI\_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty call.* (Lean: avoid the word; show the parsed-pill instead.)*

150 6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for A4? *Owner positioning call.*

151

152 ## 9. Definition of done

153 - [] From a finished job, a user can list rallies, query/select, and download a reel ? on **existing** endpoints.

154 - [] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are shown; fabricated `server`/`receiver`/`events` are hidden.

155 - [x] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens, shows a visible parsed-pill, and is unit-tested. *(P3)*

156 - [x] The P3 path is **observable**: structured, correlation-id'd logs for parse / apply / skip / empty / unavailable-request / manual-edit; unsupported concepts and zero-result queries are logged loudly, never dropped silently (§4.4). *(P3)*

157 - [x] Selection hook is unit-tested; the `min\_shot\_count` floor is surfaced, not hidden. *(P2 hook · P4 floor warning ? `lib/floor.ts` mirrors `main.py:362-383`, exempts no-shot_count rallies, blocks Build when all-dropped)*

158 - [x] From a finished job, a user can list rallies, query/select, and download a reel ? on **existing** endpoints. *(P5 ? `?video=<id>` loads via `POST /api/query`; Build ? `POST /api/compile` ? `<video>` preview + download over `GET /api/highlights/...`; verified e2e against a stub engine)*

159 - [x] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone ? the SPA **supersedes** the static `app.js` for this flow (reads `play\_segments`).

160 - [x] S7 tracker + this DoD updated. *(Doc stays live for the owner-gated M1?M5 engine milestones; not archived ? only the MVP/M0 slice is complete.)*

161

162 ## 10. References

163 **Engine code anchors `[verified this session]`**: **`backend/main.py:121-123` (config), `:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile + `min\_shot\_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135` (play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475` (query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485` (honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198` (fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe + entripoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json` (`default\_segmenter: "gemini")`).

164 **Engine docs**: **`UI\_PROPOSAL.md:46-47, 66`; `UI\_ALPHA\_EXECUTION\_PLAN.md:38-39`; `HOSTING\_PLAN.md:42`; `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md:54-55` (conversational seam built now, model punted). **This repo**: **`web/README.md`, `REQUEST\_ACCESS\_FORM.md`.

165

166 ---

167 ### Changelog

168 - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid / RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact anchor re-verified against source before writing.

169 - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON · preview not a ship-blocker · job-tethered MVP · warn-don't-override the floor); status `DRAFT` ? IN PROGRESS`; P0 done.

170 - 2026-06-20 ? **UI**: lead with rally LENGTH, park shot-count as experimental** (owner). Rally cards emphasize duration (`m:ss long`); the shot-count chip is flagged amber "experimental" with a tooltip ("under development, may be inaccurate ? filter by length instead"); example chips are duration-led (dropped "at least 8 shots"); a note states "rally length is fully supported, shot-count is experimental". `parseQuery` still accepts shot-count queries (not broken) but they're de-featured. Public badge `P5`?`alpha`. A test pins that every example chip parses to a recognized, fully-available (non-shot-count) query. Suite 75 green.

171 - 2026-06-20 ? **P5 shipped ? MVP / M0 COMPLETE**. **`App` now loads a real session from `?video=<id>` (`POST /api/query`, default ALL-selected D8) with a mode banner (demo / loading / loaded N / error); **Build** ? `compileReel` (`POST /api/compile`) ? `CompileResult` previews the stitched reel in `<video>` over `GET /api/highlights/{video\_id}/{filename}` (client-built URL ? engine returns a filepath, not a URL) + a download link; `reelFilename()` slugifies to an engine-safe bare `.mp4`. Loading/building/done/error states, all traced (`rallies.load\_\*`, `build.start/done/failed`). Verified e2e against a stub engine: load 5 rallies ? real `/api/config` floor warning ? Build ? `<video>` reached `readyState=4` (decoded a real mp4

through the proxy) + download link correct; all-below-floor selection keeps Build disabled (P4 guard prevents the 400). `reelFilename` unit-tested; suite **73 green**, typecheck + build clean. **Next: GCP-GPU end-to-end validation (real video ? rallies ? reels).**

172 - 2026-06-20 ? **P4 shipped**: `SelectionFooter` (live summary + reel-name input + Build) with an `honest` `min\_shot\_count` warning ? `lib/floor.ts` mirrors the engine floor (`main.py:362-383`) EXACTLY: a rally is dropped only if it has a numeric `shot\_count` below the floor; rallies `without` `shot\_count` are exempt; **all-below ? Build disabled** (the engine would 400). Config read from `GET /api/config` on mount (graceful offline). 8 floor unit tests; amber/red warning states verified live in-browser; suite **70 green**, typecheck + build clean.

173 - 2026-06-20 ? **P3 shipped**: deterministic `parseQuery()` (`web/src/lib/query.ts`) over length/count/order/shot_count + `runQuery`/`sortRallies`/label helpers and the `QueryBar` (live parsed-pill, example chips, greyed "not available yet" for shot-type/player/score ? never faked). Writes the shared selection set via a new `useRallySelection().apply`; `RallyGrid` ordinals made stable so a re-sort doesn't renumber cards. **First-class observability (\$4.4):** structured, correlation-id'd `query.\*` / `selection.\*` / `rally.toggle` runlog (loud `empty\_result`/`apply\_failed`; `unavailable\_request` demand signal). **Adversarial multi-agent review** (grammar/honesty/robustness/reuse) ran pre-merge ? 13 confirmed findings folded in, incl. 4 majors: an honesty leak (`at least 8 smashes` ? fake `length` ? 8s), a selection-wiping `top 0`, a unit-blind `first 30 seconds`, and self-contradictory pill labels (`first 10` shown as "top 10"). Live browser CUJ replayed (filter / top-N / honest-greying / untick / no-op / wipe-closed). Suite **62 green**, typecheck + build clean.

174

4. user (2026-06-22T09:32:54.832Z)

50b2bd7 chore(repo): ignore local runtime artifacts + correct stale NEXT_STEPS claims (#94)

---BRANCH---

master

---STATUS---

5. assistant (2026-06-22T09:32:58.777Z)

Now let me verify the engine-side claims. The doc cites many engine repo files (`badminton-highlight-indexer`). Let me check both the khelsutra repo structure and the engine files. First, let me look at the khelsutra repo's `web/` directory and the cited files.

6. user (2026-06-22T09:33:01.079Z)

=== web/ files ===

web/.gitignore

web/README.md

web/index.html

web/package-lock.json

web/package.json

web/src/App.tsx

web/src/api/client.test.ts

web/src/api/client.ts

web/src/api/locale.test.ts

web/src/api/types.ts

web/src/api/upload.test.ts

web/src/components/CompileResult.tsx

web/src/components/IngestPanel.test.tsx

web/src/components/IngestPanel.tsx

web/src/components/LanguageSwitcher.tsx

web/src/components/QueryBar.examples.test.ts

web/src/components/QueryBar.test.tsx

web/src/components/QueryBar.tsx

web/src/components/RallyGrid.tsx

web/src/components/SelectionFooter.test.tsx

web/src/components/SelectionFooter.tsx

web/src/components/SetupWizard.test.tsx

web/src/components/SetupWizard.tsx

web/src/components/StatusChip.tsx

web/src/components/TapFilters.test.tsx


```
web/src/components/TapFilters.tsx
web/src/components/a4.touch.test.tsx
web/src/components/a8.status.test.tsx
web/src/components/build-error.test.tsx
web/src/components/upload.test.tsx
web/src/hooks/useIngestJob.test.ts
web/src/hooks/useIngestJob.ts
web/src/hooks/useRallySelection.ts
web/src/hooks/useUpload.test.ts
web/src/hooks/useUpload.ts
web/src/i18n/catalog-invariants.test.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
web/src/index.css
web/src/lib/errorText.test.ts
web/src/lib/errorText.ts
web/src/lib/errors.test.ts
web/src/lib/errors.ts
web/src/lib/floor.test.ts
web/src/lib/floor.ts
web/src/lib/format.test.ts
web/src/lib/format.ts
web/src/lib/log.ts
web/src/lib/query.test.ts
web/src/lib/query.ts
web/src/lib/selection.test.ts
web/src/lib/selection.ts
web/src/main.tsx
web/src/test/setup.ts
web/src/vite-env.d.ts
web/tsconfig.json
web/vite.config.ts
=== docs/README.md exists? ===
NEXT_STEPS.md
docs/README.md
web/README.md
=== archives ===
docs/archives/FRIENDLY_AND_MULTILINGUAL.md
```

7. user (2026-06-22T09:33:02.731Z)

```
=== git log P0-P5 PRs (#11 #14 #21) ===
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
05915f2 docs(next-steps): per-rally MVP/M0 COMPLETE (P1?P5, #28/#30/#31); CUJ screencast + M1?M5
next (#32)
8bf200d feat(web): per-rally MVP P5 ? load real rallies + Build?compile?preview?download (MVP
complete) (#31)
aff8de8 feat(web): per-rally MVP P4 ? SelectionFooter + honest min_shot_count warning (#30)
66805ab docs(next-steps): per-rally P3 (parseQuery query bar) merged (#28); P4 next (#29)
d537d19 feat(web): per-rally MVP P3 ? deterministic parseQuery() query bar (traced +
adversarially hardened) (#28)
62598e7 feat(web): per-rally MVP P2 ? selectable rally grid + selection hook (unit-tested) (#21)
aac7e0c feat(web): per-rally MVP P1 ? SPA scaffold + typed engine API client (#14)
cb74ce5 docs(web): tracked design for interactive per-rally selection (RallyPicker) (#11)
```

8. assistant (2026-06-22T09:33:06.519Z)

REQUEST_ACCESS_FORM.md is referenced but not in the ls-files output. Let me check. Also note the PR numbers in the doc (P3 = "this PR", P4/P5 = "this PR") differ from actual merge PRs (#28, #30, #31). Let me verify key claims.

9. user (2026-06-22T09:33:08.681Z)

```
=== REQUEST_ACCESS_FORM ===
docs/REQUEST_ACCESS_FORM.md
=== all docs/ md ===
docs/ANNOTATE_AND_TRAIN.md
docs/DRY_RUN_CUJ.md
docs/LOCAL_APPLIANCE_ARCHITECTURE.md
docs/MULTILINGUAL_EXPANSION.md
docs/PER_RALLY_SELECTION.md
docs/PORTABILITY.md
docs/README.md
docs/REQUEST_ACCESS_FORM.md
docs/archives/FRIENDLY_AND_MULTILINGUAL.md
```

10. user (2026-06-22T09:33:26.199Z)

```
=== test count via vitest ===
{"ts":"2026-06-22T09:33:25.731Z","level":"error","event":"api.network_error","cid":"88a9ed35-3872-4029-91f8-03d331ef5f4f","method":"GET","url":"/api/capabilities","error":"Error: no engine in test"}
{"ts":"2026-06-22T09:33:25.731Z","level":"error","event":"api.network_error","cid":"82a398ee-9890-4e88-a698-f10142b62d8c","method":"GET","url":"/api/config","error":"Error: no engine in test"}
{"ts":"2026-06-22T09:33:25.783Z","level":"error","event":"api.network_error","cid":"4b3360e0-a39d-4a22-85f5-dcb52b16210d","method":"GET","url":"/api/capabilities","error":"Error: no engine in test"}
{"ts":"2026-06-22T09:33:25.783Z","level":"error","event":"api.network_error","cid":"08e28f7a-ee35-4422-8b9d-6cc41755a1e3","method":"GET","url":"/api/config","error":"Error: no engine in test"}
{"ts":"2026-06-22T09:33:25.783Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

```
stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-22T09:33:25.783Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

```
stderr | src/i18n/i18n.test.tsx > MULTILINGUAL Phase A (es/da/id) >
renders each new locale's catalog (real translations, not English / not raw keys)
{"ts":"2026-06-22T09:33:25.784Z","level":"warn","event":"wizard.capabilities_unavailable","cid":"4b3360e0-a39d-4a22-85f5-dcb52b16210d","error":"ApiError: Network error calling GET /api/capabilities"}
{"ts":"2026-06-22T09:33:25.784Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

```
....■[
32m..
```

```
Test Files 24 passed (24)
Tests 223 passed (223)
Start at 15:03:14
Duration 11.62s (transform 12.60s, setup 430ms, collect 27.69s, tests 10.49s,
environment 85.97s, prepare 10.58s)
```

11. assistant (2026-06-22T09:33:32.274Z)

Test suite is now 223 tests (doc claims 73/75 at MVP completion ? that was accurate at the time, but the doc doesn't make a current-total claim except in the changelog, which is historical). REQUEST_ACCESS_FORM is at `docs/REQUEST\_ACCESS\_FORM.md`, but the doc references it as `REQUEST\_ACCESS\_FORM.md` (repo-root-style) in §0.6 and §10. Let me check the exact references and verify the engine-side anchors.

Let me verify the engine repo file paths/line anchors. The doc is heavy on `[verified backend/main.py:... ]` claims.

12. user (2026-06-22T09:33:34.748Z)

```
=== engine HEAD ===
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
=== main.py exists? ===
-rw-r--r-- 1 avidu 197609 58163 Jun 22 14:52 backend/main.py
=== route greps ===
297:@app.get("/api/config")
307:@app.get("/api/health")
374:@app.get("/api/videos")
611:@app.get("/api/jobs/{job_id}/cost")
626:@app.get("/api/videos/{video_id}/cost")
637:@app.get("/api/jobs/{job_id}")
670:@app.get("/api/highlights/{video_id}/{filename}")
694:@app.post("/api/query")
705:@app.post("/api/compile")
```

13. assistant (2026-06-22T09:33:40.986Z)

The Bash tool's cwd resolved against the engine repo? No ? the env says working dir is badminton-highlight-indexer. The Bash `cd` resets between calls. The previous command had no `cd`, so it ran in the default dir which is `C:\Users\avidu\Projects\badminton-highlight-indexer` (the engine repo). That's exactly the engine repo I need.

Important findings: the engine `main.py` line numbers have drifted significantly. The doc claims `/api/config` at `:121-123`, but it's now at `:297`. `/api/health` is claimed as "new" (P6, ? not done) but it **exists** at `:307`. `/api/videos` is claimed as "new" (P8, ? gated, ?) but it **exists** at `:374`. Let me dig into these.

14. user (2026-06-22T09:33:42.469Z)

```
=== /api/health body ===
@app.get("/api/health")
def health():
    """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
    (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
    confirm the engine is reachable and which detector is wired. Never raises."""
    sport = getattr(global_config, "sport", None)
    indexing = getattr(global_config, "indexing", None)
    return {
        "status": "ok",
        "sport": sport,
        "default_segmenter": getattr(indexing, "default_segmenter", None),
    }
```

P1: the bundled SPA pins this ? the contract version of the /api surface. BUMP when an endpoint's request/response shape changes incompatibly (purely additive endpoints)

```
API_VERSION = 1
```

```
def _engine_version() -> str:
    """Installed package version (wheel/editable); 'unknown' if not importable as a
    distribution."""
    try:
        return _pkg_version("badminton-highlight-indexer")
    except PackageNotFoundError:
        return "unknown"
=== /api/videos body ===
@app.get("/api/videos")
def list_videos(limit: Optional[int] = None):
    """List indexed videos (most recent first) so the console survives a reload ? previously
    only a
    single-row get existed (PACKAGING_PLAN.md P1)."""
```

```

    return {"videos": db_instance.list_videos(limit=limit)}

def _reels_dir() -> str:
    return getattr(getattr(global_config, "output", None), "output_dir", "output") or "output"

_REEL_EXTS = (".mp4", ".mov")

@app.get("/api/reels")
def list_reels():
    """List compiled highlight reels present in the output dir, newest first, with a download
    URL.

    NB: reels are stored FLAT in output_dir under the bare name passed to /api/compile ? the
    engine
    does NOT track a video_id?reel association today (a per-video listing would be a phantom),
    so
    this is a global list. (Per-video reel tracking would need a DB record + compile change ? a
    later slice.) Only top-level video files are listed (per-video scratch subdirs are
    skipped)."""
    base = _reels_dir()
    reels = []
    try:
        entries = os.scandir(base)
    except OSError:
        return {"reels": []}
    with entries:
        for e in entries:
            if e.is_file() and e.name.lower().endswith(_REEL_EXTS):
                try:
                    st = e.stat()
                except OSError:
                    continue
                reels.append({
                    "filename": e.name,
                    "size_bytes": st.st_size,
                    "modified_at": st.st_mtime,
                    "download_url": f"/api/reels/{quote(e.name)}",
                })
    reels.sort(key=lambda r: r["modified_at"], reverse=True)
    return {"reels": reels}

@app.get("/api/reels/{filename}")
def download_reel(filename: str):

```

15. assistant (2026-06-22T09:33:48.211Z)

Confirmed: `/api/health`` and `/api/videos`` now both exist in the engine, but the doc's §4.3 table and §7 tracker mark them as `***new***` / P6 & P8 (? , not done). This is staleness ? the engine shipped these endpoints. Note: these are engine-repo deliverables (P6-P10), separate from the khelsutra SPA work this doc's MVP is about. The doc explicitly notes P6-P10 are `“(engine repo)”`. Let me verify the remaining endpoint anchors and the compile/query/config anchors to assess what's actually drifted vs the doc's "exists" claims.

16. user (2026-06-22T09:33:49.964Z)

```

=== query @694 ===
@app.post("/api/query")
def query_play_segments(req: QueryRequest):
    filters = {
        "server": req.server,
        "receiver": req.receiver,
    }

```

```

        "duration_min": req.duration_min,
        "event_type": req.event_type
    }
    segments = db_instance.query_play_segments(req.video_id, filters)
    return {"play_segments": segments}

@app.post("/api/compile")
=== compile @705 head ===
@app.post("/api/compile")
def compile_highlights(req: CompileRequest):
    video = db_instance.get_video(req.video_id)
    if not video:
        raise HTTPException(status_code=404, detail="Indexed video record not found.")

    # Retrieve matching play segments from db
    all_segments = db_instance.query_play_segments(req.video_id, {})
    matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]

    if not matched_segments:
        raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")

    # Reject path traversal / absolute output names (os.path.join would otherwise let an
    # absolute or '..'-laden output_filename write anywhere on disk).
    try:
        out_name = safe_output_filename(req.output_filename)
    except ValueError as e:
        raise HTTPException(status_code=400, detail=str(e)) from e

    # The configured shot-count floor (stitching.min_shot_count) applies on the API
    # path too ? parity with scripts/run_process_and_stitch.py. Segments WITHOUT shot_count
    # metadata are exempt: the floor filters known counts only, so explicitly
    # requested manual/legacy segments can't silently vanish under the default floor.
    stitching = getattr(global_config, "stitching", None) or StitchingConfig()
    kept = []
=== config @297 ===
@app.get("/api/config")
def get_config():
    # P0b: this endpoint is unauthenticated, so never serialize a credential-shaped field. Today
    # AppConfig holds no secrets (the Gemini key is env-only; cf_access_aud is a public id), so
    this
    # is byte-identical for the real fields ? but it fails-closed if a secret is ever added
    (guarded
    # by tests/test_redact.py + the /api/config redaction test).
    if global_config is None:
        return {}
    return redact_secrets(global_config.model_dump(mode="json"))

@app.get("/api/health")
=== jobs @637 ===
@app.get("/api/jobs/{job_id}")
def get_job(job_id: str):
    """Job state: {status, progress, result, reports}. `status` = execution state
    (queued|running|done|failed); the pipeline verdict is result["status"]."""
    job = db_instance.get_job(job_id)
    if not job:
        raise HTTPException(status_code=404, detail=f"Unknown job_id '{job_id}'")
    result = job.get("result")
    return {
        "job_id": job["id"],
        "status": job["status"],
        "progress": job.get("progress"),
        "video_id": job.get("video_id"),
        "error": job.get("error"),
        "result": result,
    }

```

```

    "reports": (result or {}).get("reports"),
    "created_at": job.get("created_at"),
    "started_at": job.get("started_at"),
    "finished_at": job.get("finished_at"),
}

```

--- Chunked upload (B2, PR-2) -----

The chunked-upload island (in-process registry + helpers + the 4 endpoints) now lives backend.api.uploads as a self-contained APIRouter. It resolves the live startup config

17. assistant (2026-06-22T09:33:56.890Z)

The core MVP endpoints (`/api/query`, `/api/compile`, `/api/config`, `/api/jobs`, `/api/highlights`) all still **exist** with the documented shapes ? so the doc's central "MVP runs on existing endpoints" thesis remains correct, even though the exact line numbers have drifted. The line-number drift is in `[verified ...]` anchors throughout.

Now the more material staleness: the doc's §4.3 table marks `/api/health` and `/api/videos` as **"new"** (not yet built), and §7 tracker marks P6 (`/api/health`) and P8 (`/api/videos`) as ? (todo). Both endpoints now exist in the engine. Let me confirm `/api/jobs` returns `video\_id` (the doc says `result.video\_id`; actual returns top-level `video\_id` plus `result`). Let me also verify the static `app.js` bug claim and the `default\_segmenter` default, plus check whether the doc's MVP DoD items (P1-P5) genuinely shipped in the SPA.

18. user (2026-06-22T09:33:59.307Z)

```

=== static app.js exists & data.rallies? ===
backend/api/static/app.js
138:         // Successfully processed (field is play_segments ? the old `rallies` read was a
bug)
139:         indexedRallies = data.play_segments || [];
344:         renderRallyCards(data.rallies || []);
=== default_segmenter in config.json ===
26:     "default_segmenter": "gemini",
=== compile return (download_url? filepath?) ===
30:     # The stored `filepath` is the SOURCE ref ? a bucket/Drive URI (gs://, s3://,
gdrive://) for a
35:     local_src = storage.acquire_local_input(video["filepath"], getattr(global_config,
"storage", None))
37:     # `filepath` is server-local (not reachable by the browser); also return the ready-
to-use
39:     download_url = f"/api/highlights/{quote(req.video_id)}/{quote(out_name)}"
40:     return {"status": "success", "filepath": out_path, "download_url": download_url}

```

19. assistant (2026-06-22T09:34:06.862Z)

Significant finding: `/api/compile` now **returns `download\_url`** (line ~769). The doc repeatedly claims compile "returns a **server-local filepath**, NOT a URL" and that the SPA must "construct this download URL itself" (§0, §3, §4.2, §4.3 P6 "modify", D11 risk table, changelog). That is now stale ? the engine adds `download\_url` (this was P6's "modify" item, which the doc marks as ? not-done). And `/api/health` exists (P6's "new" item).

So **all of P6's engine work** (`/api/health` + compile `download\_url`) is shipped, and P8's `/api/videos` is shipped ? but the doc's §4.3 and §7 mark them as not-built. Let me confirm the `default\_segmenter: "gemini"` claim (verified ? line 26) and the gemini.py honest-fields claim quickly, then check the `docs/README.md` / `NEXT\_STEPS.md` pointer claims in khelsutra.

20. user (2026-06-22T09:34:09.205Z)

```

=== gemini server/receiver None ===
479:         # Extract shot_count from Gemini or compute estimate if not present
480:         shot_count = segment.get("shuttle_exchange_count")

```

```

481:         if shot_count is None:
482:             shot_count = max(3, int(dur / 0.8)) # 1 shot every 0.8 seconds as fallback
estimate
485:             shot_count = int(shot_count)
487:             shot_count = max(3, int(dur / 0.8))
491:             "shot_count": shot_count,
492:             "shot_count_confidence": segment.get("shot_count_confidence", "Unknown"),
=== motion random fields ===
3:import random
181:     For each detected segment this assigns a random server/receiver, fabricates
183:     with the stdlib ``random`` module (the legacy demo's fabricated baseline).
184:     Returns the per-segment dicts handed back to the caller. The ``random``
190:     # Assign Server and Receiver randomly from pool
191:     server, receiver = random.sample(player_pool, 2)
196:     "shot_count": random.randint(4, 25),
198:     "avg_speed_kmh": round(random.uniform(40, 85), 1)
206:     serve_foul = random.random() < 0.15 # 15% chance of serve foul
212:     details={"status": "foul" if serve_foul else "valid", "type":
"low_serve" if random.random() > 0.5 else "high_serve"}

```

21. user (2026-06-22T09:34:11.661Z)

```

=== README pointer to PER_RALLY ===
docs/README.md:22: architecture. Umbrella for the console; per-rally selection is one screen of
it. `DRAFT`.
docs/README.md:23:- PER_RALLY_SELECTION.md ? interactive pick-rallies ? reel builder
(RallyPicker). The alpha
NEXT_STEPS.md:29: (Playwright recording of the per-rally SPA driven against real GPU-run data ?
`promo_cuj.mp4`/`.gif`
NEXT_STEPS.md:32:- Interactive per-rally selection (big product feature) ? ? MVP / M0
COMPLETE (the full
NEXT_STEPS.md:34: (`web/`). Tracked design + P-tracker (source of truth):
`docs/PER_RALLY_SELECTION.md` (IN PROGRESS;
NEXT_STEPS.md:39: (P6), RESTful `?/rallies` alias (P7), multi-video home + auth (P8), per-rally
preview/timeline (P9),
NEXT_STEPS.md:59:- Alpha SPA in `web/` (Vite + React + TS + Tailwind) ? hosts the operator
console + the per-rally selection UI above.
=== DRY_RUN_CUJ $9 ===
1:# Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local (multilingual)
8:Operate the web UI ? point at a video on the GCS bucket ? the engine (tool) processes
it ? review/select rallies in the UI ? compile ? download the reel to the local
machine. Then repeat with the UI in Hindi to prove the multilingual workflow.
14:- The SPA has no ingest/process UI ? it can't point at a video (bucket or local), can't
call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in the
URL. This SPA front-half is the only net-new work for Path B (= B-P2).
15:- HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified
file:line]` POST /api/process accepts a `gs://`/`gcs://` (also `s3://`/`local://`)
`video_path` directly. Submit resolves the URI and skips the local-existence check for non-
local backends (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`/`gcs`
normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch
(`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the same
hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into
the API's SQLite `videos`/`play_segments`. Proven end-to-end by
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202,
not 404 ? job `done` ? `db.get_video(video_id)` row with the source URI preserved), plus the
hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
16: - Where the earlier "net-new" claim went wrong: it conflated the API path
(`/api/process`, which DID gain URI ingest in M2) with the separate worker CLI (`python -m
backend.worker infer --video-uri gcs://`), which writes `intervals.csv`/`report.json` to the
bucket ? not the API DB. The CLI distinction is real; the "the endpoint is local-only"
conclusion was not. ? Engine-surface claims drift between sessions (the engine ships PRs
independently) ? re-verify any `[gap]`/`net-new` engine claim against the cited `file:line`
before scoping it as work.
19:Path A ? Cheapest dry run NOW (~zero new code). Pre-process the bucket video out-of-band
to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the

```

****multilingual UI-driven workflow**** end-to-end. The only non-UI step is "point at the bucket" (a one-line CLI pre-process).

21: ****Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first scoped).****
 The ****HTTP bucket-ingest endpoint already exists**** (§1) ? so Path B is ****only the SPA front-half**** (appliance S1/S2): an ****ingest**** input (enter a `gs://` URI) ? `POST /api/process` ? an ****ingest + progress**** screen polling `GET /api/jobs/{id}` ? on `done`, load `result.video\_id` (which the existing back-half already renders). ****No net-new engine endpoint**** ? B-P1 is already done; the remaining work is ****B-P2 (the SPA)****.

25:2. ****Get the bucket video local + into the API DB:****

26: `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST /api/process {"video\_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET /api/jobs/{job\_id}` until `result.video\_id`. *(`gemini` needs `GEMINI\_API\_KEY`; `motion` is key-free but hides fabricated metadata.)*

34:1. ****Scope:**** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? ****SPA-only****; the engine endpoint already accepts `gs://`, see §1)? ***Recommend A first*** (proves the multilingual workflow fast), then B as the appliance build. ? ? ****RESOLVED: A first (done, §7), then B (B-P2, the SPA ingest/progress screen).****

22. assistant (2026-06-22T09:34:18.567Z)

The DRY_RUN_CUJ.md doc structure differs from what PER_RALLY's status header claims. The PER_RALLY header says: "Validated live e2e against a REAL engine (Gemini) via the in-UI bucket?reel CUJ ? `DRY\_RUN\_CUJ.md` §9, 2026-06-21". But DRY_RUN_CUJ.md §9 ? let me check whether a §9 exists and whether it documents a real-engine e2e validation. The grep showed §1 content. Let me check the section structure of DRY_RUN_CUJ.md.

23. user (2026-06-22T09:34:21.212Z)

```
=== section headers in DRY_RUN_CUJ ===
1:# Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local (multilingual)
7:## 0. The goal (owner's words)
10:## 1. What already works vs the gap `[verified]`
18:## 2. Two paths to the goal
23:## 3. Path A runbook (the cheapest dry run) `[verified-steps]`
33:## 4. Open decisions ? OWNER (these gate the dry run)
39:## 5. Honest caveats
46:## 6. Definition of done (the dry run)
52:## 7. ? Run result ? Path A, local (2026-06-21)
62:### ? Fix the dry run surfaced (engine) ? MERGED-PENDING
65:## 8. ? Correction (2026-06-21): B-P1 (the bucket-ingest endpoint) was already shipped ?
verified against code
80:## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)
87:### ? Bug the run surfaced + fixed (engine) ? MERGED
90:### Changelog
=== count ## sections ===
10
```

24. assistant (2026-06-22T09:34:26.875Z)

DRY_RUN_CUJ.md §9 does exist and is "? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)". So the PER_RALLY header's claim "Validated live e2e against a REAL engine (Gemini) via the in-UI bucket?reel CUJ ? `DRY\_RUN\_CUJ.md` §9" is ****correct**** (verified the cross-reference resolves to a real, matching section).

Now let me verify the PR-number claims in §7 tracker. The doc lists: P0=#11, P1=#14, P2=#21, and P3/P4/P5 = "(this PR)". From the git log, actual merges were P3=#28, P4=#30, P5=#31. The "(this PR)" placeholders were never updated to real PR numbers ? that's a staleness item (a reader can't find the PR). Let me confirm #11/#14/#21 are correct and capture the P3/P4/P5 real numbers.

25. user (2026-06-22T09:34:28.800Z)

05915f2 docs(next-steps): per-rally MVP/M0 COMPLETE (P1?P5, #28/#30/#31); CUJ screencast + M1?M5 next (#32)


```

8bf200d feat(web): per-rally MVP P5 ? load real rallies + Build?compile?preview?download (MVP
complete) (#31)
aff8de8 feat(web): per-rally MVP P4 ? SelectionFooter + honest min_shot_count warning (#30)
66805ab docs(next-steps): per-rally P3 (parseQuery query bar) merged (#28); P4 next (#29)
d537d19 feat(web): per-rally MVP P3 ? deterministic parseQuery() query bar (traced +
adversarially hardened) (#28)
62598e7 feat(web): per-rally MVP P2 ? selectable rally grid + selection hook (unit-tested) (#21)
aac7e0c feat(web): per-rally MVP P1 ? SPA scaffold + typed engine API client (#14)
cb74ce5 docs(web): tracked design for interactive per-rally selection (RallyPicker) (#11)
=== confirm #11 #14 #21 messages ===
62598e7 feat(web): per-rally MVP P2 ? selectable rally grid + selection hook (unit-tested) (#21)
aac7e0c feat(web): per-rally MVP P1 ? SPA scaffold + typed engine API client (#14)
cb74ce5 docs(web): tracked design for interactive per-rally selection (RallyPicker) (#11)

```

26. assistant (2026-06-22T09:34:42.128Z)

Confirmed. P0=#11, P1=#14, P2=#21 are correct. P3=#28, P4=#30, P5=#31 actually merged but the doc shows "(this PR)" placeholders ? never backfilled.

Now let me verify the SPA-side P2-P5 deliverables genuinely shipped (the files exist ? confirmed earlier: `QueryBar.tsx`, `SelectionFooter.tsx`, `CompileResult.tsx`, `useRallySelection.ts`, `floor.ts`, `query.ts`). And verify the `web/README.md` "Vite + React + TS + Tailwind" claim and the engine PATCH/segments and other "new" P9/P10 endpoint claims (still genuinely not built). Let me do a final batch.

27. user (2026-06-22T09:34:44.319Z)

```

=== web stack ===
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview",
  "test": "vitest run",
  "test:coverage": "vitest run --coverage"
  "react": "^19.0.0",
  "react-dom": "^19.0.0",
  "react-i18next": "^17.0.8"
  "@tailwindcss/vite": "^4.0.0",
  "@testing-library/react": "^16.3.2",
=== Tailwind present? ===
web/package.json:   "@tailwindcss/vite": "^4.0.0",
web/package.json:   "tailwindcss": "^4.0.0",
web/vite.config.ts:import tailwindcss from "@tailwindcss/vite";
web/vite.config.ts:  plugins: [react(), tailwindcss()],
web/vite.config.ts:import tailwindcss from "@tailwindcss/vite";
web/vite.config.ts:  plugins: [react(), tailwindcss()],

```

28. user (2026-06-22T09:34:46.410Z)

```

=== PATCH /api/segments exists? (should NOT for P10) ===
237:# this module object at call-time, so `monkeypatch.setattr(backend.main, ...)` still
238:# intercepts the inter-function calls (a parked job arms the PATCHED `_arm_wake_timer`,
1114:  # `None.claim_due_job()` and every submitted job hangs in 'queued' forever. Dispatch
through
=== /api/videos/{id}/rallies alias exists? (P7) ===
=== source/clip/thumbnail (P9)? ===
197:# absent (a source checkout with no vendored UI). The SPA calls same-origin `/api`, so no
SPA change
456:  # req.video_path (the source URI) is kept for the DB record + report; only the file-
reading
613:  """"Per-job cost (COMPUTE_DECOUPLED_SERVING M8, D8). ``cost_usd`` is the source of truth
(matches
774:def run_cli_diagnose_clip(video_path: str, timestamp: float):
864:  parser.add_argument("--output-dir", default="output", help="Directory where processed
clips/highlights are saved")

```

```

866:     parser.add_argument("--debug-clip", type=float, help="Timestamp (in seconds) of a video
clip to run detailed diagnostics on")
919:     # clip carries the same configured stitching paddings + watermark as a real
931:     if args.debug_clip is not None:
933:         print("[ERROR] Please provide --video-path to debug a specific clip.")
935:         run_cli_diagnose_clip(args.video_path, args.debug_clip)

```

29. assistant (2026-06-22T09:34:53.756Z)

Confirmed:

- Web stack: React + Tailwind + Vite + Vitest all present. Correct.
- P7 (`/api/videos/{id}/rallies` RESTful alias): does NOT exist ? accurately marked ?.
- P9 (source/clip/thumbnail stream): does NOT exist ? accurately marked ?.
- P10 (`PATCH /api/segments/{id}`): does NOT exist ? accurately marked ?.

So the doc's P7/P9/P10 "not built" status is **correct**. The staleness is concentrated in **P6** (both items shipped: `/api/health` exists + compile returns `download\_url`) and P8 (`/api/videos` exists) ? these are marked as not-built but are now built.

Let me verify the P8 `/api/videos` and the "only `get\_video(id)` exists, `database.py:418`" claim ? the doc says no list-videos endpoint and no DB enumerate. We saw `db\_instance.list\_videos` is now called. Let me confirm the DB method exists, completing the P8 contradiction.

30. user (2026-06-22T09:34:55.565Z)

```

=== list_videos in database.py? ===
536:     def get_video(self, video_id: str) -> Optional[Dict[str, Any]]:
545:     def list_videos(self, limit: Optional[int] = None) -> List[Dict[str, Any]]:
774:     def get_video_cost(self, video_id: str) -> Dict[str, Any]:
=== play_segments schema cols ===
95:         """Initializes tables for videos, play_segments, and events."""
172:         """Create the videos / play_segments / events base tables (idempotent)."""
186:         CREATE TABLE IF NOT EXISTS play_segments (
189:             start_time REAL NOT NULL,
190:             end_time REAL NOT NULL,
191:             duration REAL NOT NULL,
192:             server TEXT,
193:             receiver TEXT,
208:             FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE CASCADE
222:             start_time REAL NOT NULL,
223:             end_time REAL NOT NULL,
224:             duration REAL NOT NULL,
233:             FOREIGN KEY (confirmed_segment_id) REFERENCES play_segments(id) ON DELETE
SET NULL
432:             p = cur.execute("SELECT COALESCE(MAX(id), 0) FROM play_segments WHERE video_id =
?",
441:             """Delete play/candidate segments by id range (events cascade from play_segments).

```

31. assistant (2026-06-22T09:35:08.318Z)

Confirmed: `db.list\_videos()` now exists (line 545), directly contradicting the doc's claim that "only `get\_video(id)` exists, `database.py:418`" ? and the line number `:418` for `get\_video` has drifted to `:536`. The `play\_segments` schema columns (start_time, end_time, duration, server, receiver) are confirmed (line 186+), so that foundational claim is correct, though the line anchor (`:124-135`) has drifted to `~:186-200`.

I have enough evidence. Let me summarize the findings. The doc is a tracked design/progress doc whose **MVP** (P1-P5) SPA claims are accurate and shipped, but several **engine-side** status markers (P6, P8) are stale (those endpoints now exist), the **"compile returns a filepath not a URL"** claim is now false (it returns `download\_url`), the **P3/P4/P5** PR numbers were never backfilled from "(this PR)", and **most** `[verified backend/main.py:NNN]` line anchors have drifted (engine ships PRs independently ? a known hazard the sibling DRY_RUN_CUJ doc itself flags).

Let me record the structured output.

32. user (2026-06-22T09:35:50.656Z)

Structured output provided successfully

33. assistant (2026-06-22T09:36:02.999Z)

Audit complete. Summary of what I found in `docs/PER\_RALLY\_SELECTION.md`:

****Overall: significant-staleness**** (not an archive candidate ? the doc is still live for the owner-gated M1-M5 work).

The MVP thesis holds up ? the SPA-side P1-P5 deliverables genuinely shipped (verified files in `web/` + git PRs #14/#21/#28/#30/#31), and the future P7/P9/P10 engine endpoints are correctly marked not-built. But the engine moved on under the doc:

High-severity (a reader would act wrongly):

- ****P6 is shipped, doc says ?.** `/api/health`` exists at `backend/main.py:307``, and `/api/compile`` now returns `download_url`` ? both P6 items the doc lists as not-done.
- ****The core "compile returns a filepath NOT a URL" caveat is now false**** (repeated in §0, §3, §4.2, §4.3, D11). The endpoint returns both `filepath`` and `download_url``.
- ****P8's `/api/videos`` is shipped, doc says ?/"new".\*\*** Endpoint at `main.py:374``, `db.list_videos()`` at `database.py:545`` ? directly contradicting the doc's "only `get_video(id)`` exists, `database.py:418``".

Medium-severity:

- ****P3/P4/P5 PR cells still say "(this PR)"**** ? never backfilled to the real #28/#30/#31.
- ****Nearly every `[verified backend/main.py:NNN]`` / `database.py`` line anchor has drifted**** (config 121?297, jobs 274?637, query 331?694, compile 342?705, play_segments 124?186, get_video 418?536). Contracts still exist and behave as described; only the line numbers are wrong.

Correct (verified): the status-header e2e claim resolves to a real `DRY_RUN_CUJ.md §9``; the honest-fields/default-segmenter/motion-fabrication claims; the `app.js`` `data.rallies`` bug; the web stack; and P7/P9/P10 genuinely unbuilt.

Relevant files: `C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md`` (audited),
`C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py``,
`C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py``.